

1 Solution — GIAM 7.1

Problem 3

1.1 Problem

In how many ways can you get “*two pairs*” in Yahtzee?

1.2 Setup

Yahtzee is played with five standard six-sided dice, so each die shows a value in $\{1, 2, 3, 4, 5, 6\}$. Treat the dice as indistinguishable: a *hand* is the multiset of five face values one sees on the table.

A *two-pair* hand has the form

$$\{v, v, w, w, x\},$$

where v, w, x are three *distinct* values from $\{1, \dots, 6\}$. The pairs sit at values v and w ; the lone die (the “kicker”) sits at value x . Requiring $v \neq w$ rules out four-of-a-kind, and requiring $x \notin \{v, w\}$ rules out full house and five-of-a-kind. So this is the “two pair, nothing better” hand of poker fame.

1.3 Count

A first pass at the multiplication principle picks the data in three steps:

- pick the *first* pair value: 6 choices,
- pick the *second* pair value, different from the first: 5 choices,
- pick the singleton value, different from both pair values: 4 choices,

giving $6 \cdot 5 \cdot 4 = 120$. But this **over-counts by a factor of 2**: the two pair values v and w play symmetric roles in the hand $\{v, v, w, w, x\}$, so labelling them in the order (v, w) and in the order (w, v) produces the *same* multiset. Dividing,

$$\#\text{two-pair hands} = \frac{6 \cdot 5 \cdot 4}{2} = \boxed{60}.$$

Equivalently, choose the *unordered* set of two pair values out of six — $\binom{6}{2} = 15$ ways — then choose the singleton from the remaining four values — 4 ways:

$$\binom{6}{2} \cdot 4 = 15 \cdot 4 = 60.$$

1.4 Contrast with the Full-House Count

It is instructive to see *why* the full-house count (Problem 2) did **not** require dividing by 2. There the hand has the form $\{v, v, v, w, w\}$: the triple value v appears three times and the pair value w appears twice, so they play *different* roles. The two multisets

$$\{1, 1, 1, 2, 2\} \quad \text{and} \quad \{2, 2, 2, 1, 1\} = \{1, 1, 2, 2, 2\}$$

are *not* equal — they differ in which value appears three times — and so must both be counted. Hence the full-house tally is simply $6 \cdot 5 = 30$, with no over-counting.

For two pair both pairs have multiplicity 2, so the pair-values are interchangeable in the multiset, and the factor of 2 correction is necessary.

Hand type	Pattern	Symmetric values?	Naive count	Correction	Final
Full house	$\{v, v, v, w, w\}$	No	$6 \cdot 5 = 30$	$\div 1$	30
Two pair	$\{v, v, w, w, x\}$	Yes (v and w)	$6 \cdot 5 \cdot 4 = 120$	$\div 2$	60

Table 1.1

1.5 Enumeration

The 60 two-pair hands organize as a 15×4 grid indexed by the *unordered* pair-value set $\{v, w\}$ and the singleton $x \notin \{v, w\}$. For example, the row corresponding to pair-values $\{1, 2\}$ contains the four hands

$$\{1, 1, 2, 2, 3\}, \quad \{1, 1, 2, 2, 4\}, \quad \{1, 1, 2, 2, 5\}, \quad \{1, 1, 2, 2, 6\}.$$

Repeating across all $\binom{6}{2} = 15$ pair-value sets — namely

$\{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 5\}, \{1, 6\}, \{2, 3\}, \{2, 4\}, \{2, 5\}, \{2, 6\}, \{3, 4\}, \{3, 5\}, \{3, 6\}, \{4, 5\}, \{4, 6\},$

— gives $15 \cdot 4 = 60$ hands, confirming the count.

1.6 Remark — Counting *Rolls* Instead of Hands

If one instead counts *rolls* of five labeled dice — the natural sample space for probability — each multiset $\{v, v, w, w, x\}$ arises from

$$\frac{5!}{2! 2! 1!} = \frac{120}{4} = 30$$

distinct ordered five-tuples, one per arrangement of the multiset. Hence

$$\# \text{rolls producing two pair} = 60 \cdot 30 = 1800.$$

Out of $6^5 = 7776$ equally likely rolls of five dice, 1800 produce a two-pair hand, giving

$$\Pr(\text{two pair on a single throw}) = \frac{1800}{7776} = \frac{25}{108} \approx 23.15\%.$$

As a cross-check, the seven mutually exclusive single-roll “patterns” must partition the 7776 outcomes:

Pattern	Ordered rolls	Fraction
All distinct	720	$\approx 9.26\%$
One pair only	3600	$\approx 46.30\%$
Two pair	1800	$\approx 23.15\%$
Three of a kind only	1200	$\approx 15.43\%$
Full house	300	$\approx 3.86\%$
Four of a kind	150	$\approx 1.93\%$
Five of a kind	6	$\approx 0.08\%$
Total	7776	100%

Table 1.2

The totals add up — confirming both that two pair contributes **1800** rolls and that **60** is the correct unordered count.